Questions

Q1.

*Figure 28 shows a cross-section of a boat floating in water and the same boat with a load inside.

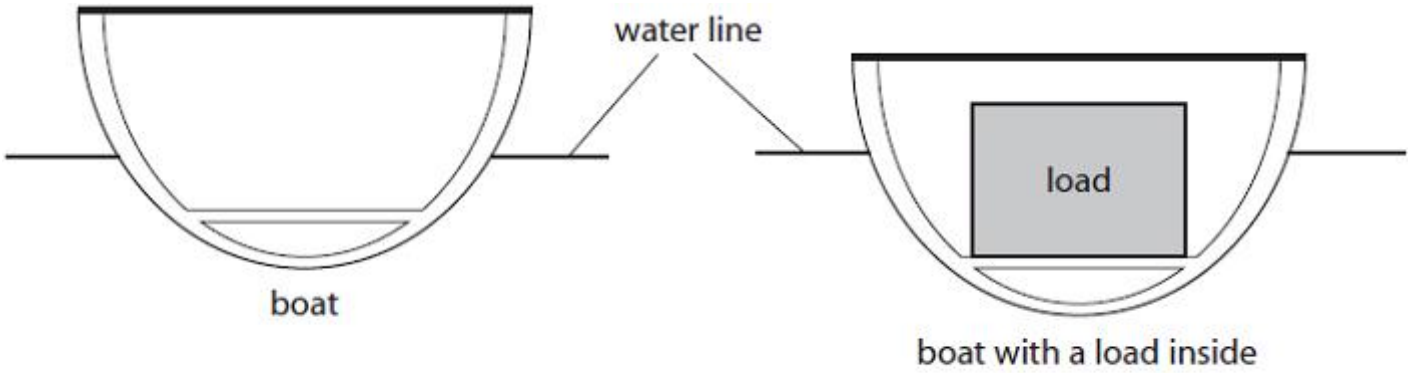

Figure 28

The boat floats lower in the water when there is a load inside the boat.

Explain why the boat floats in water and why the boat floats lower in the water when there is a load inside the boat.

You may add to the diagram to help with your answer.

(6)

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(Total for question = 6 marks)

Q2.

* Figure 26 shows the submarine stationary and submerged at a depth of 10 m.

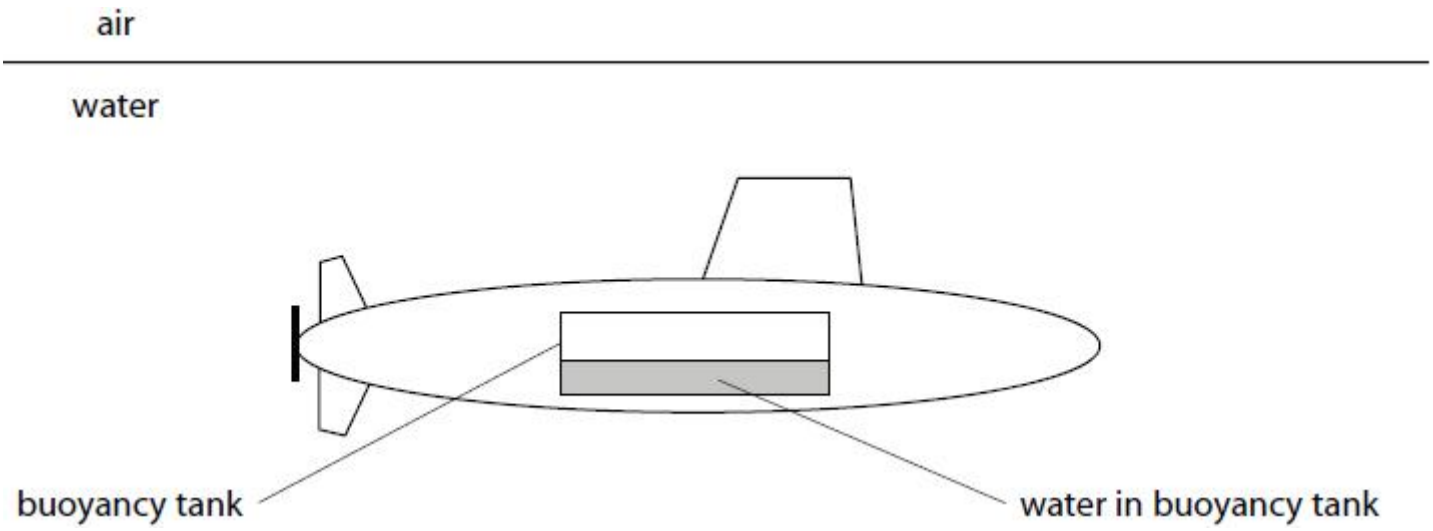

Figure 26

Explain how pumping water into and out of the buoyancy tank affects the depth of the submarine below the surface.

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(Total for question = 6 marks)

Q3.

Figure 3 shows three containers A, B, and C.

Each container contains a liquid, as shown.

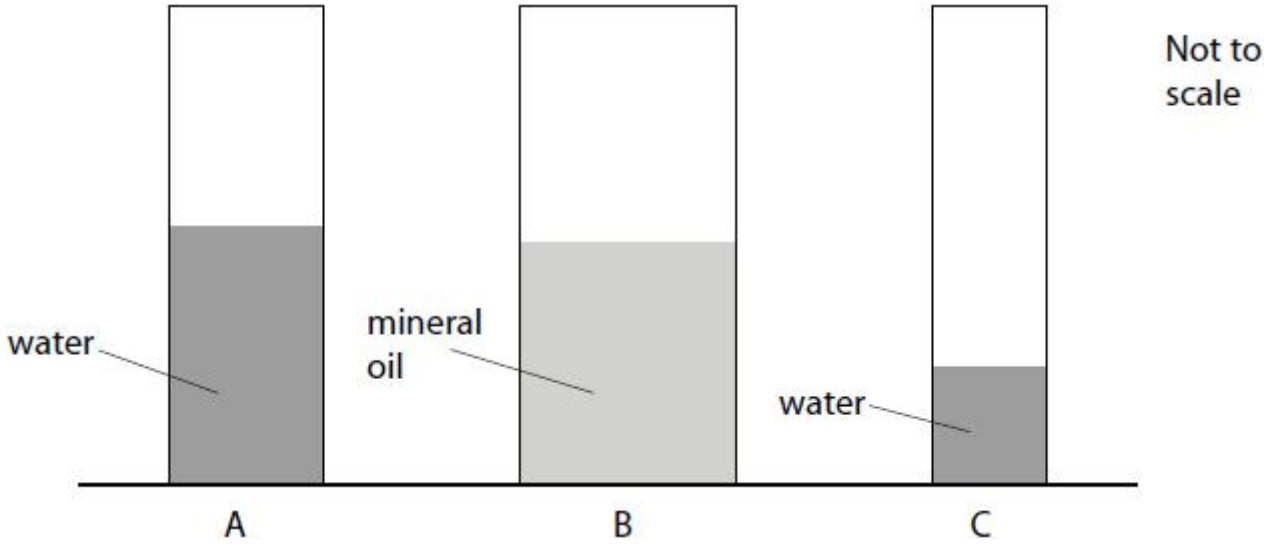

Figure 4 gives some data about the liquids and containers.

container	area of base (cm^2)	name of liquid	density of liquid (g/cm^3)	depth of liquid in container (cm)
A	16	water	1.00	50.00
B	32	mineral oil	0.91	50.00
C	12	water	1.00	25.00

Figure 4

Explain which container has the highest pressure at the bottom, and which container has the lowest.

Use information from Figure 3 and Figure 4.

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(Total for question = 3 marks)

Q4.

Figure 12 is a model representing molecules of the Earth's atmosphere.

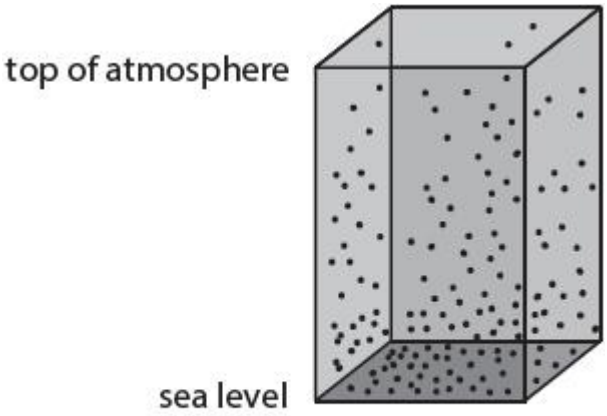

Figure 12

Use Figure 12 to explain how the density of the air varies with height above sea level.

(2)

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(Total for question = 2 marks)

Q5.

Figure 11 shows two drawings of the same person on a bed.

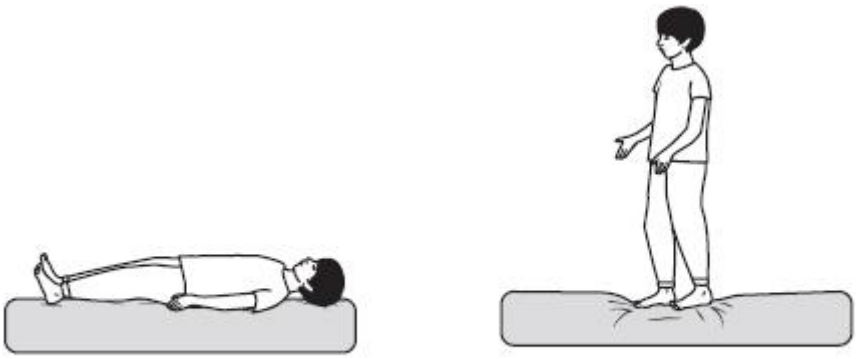

Figure 11

Explain why the person exerts a different pressure on the bed when standing up than when lying down.

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(Total for question = 2 marks)

Q6.

A student investigates the stretching of a long piece of rubber.

Figure 15 shows the apparatus to be used.

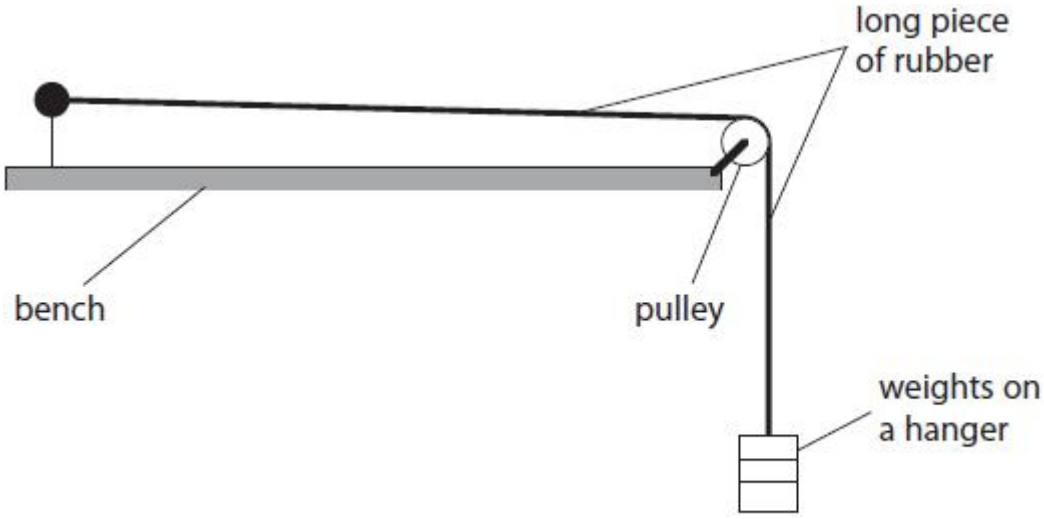

Figure 15

The student puts just enough weight on the weight hanger to make the piece of rubber just tight.

The student wants to plot a graph to show how the extension of the piece of rubber varies with the force used to stretch it.

The student adds a known weight to the weight hanger.

(i) Describe how the student could measure the extension of the rubber when he adds another weight to the weight hanger.

(2)

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(ii) The student obtains a series of values of force and extension while loading the piece of rubber and then unloading it.
Figure 16 shows the graph of the student's values.

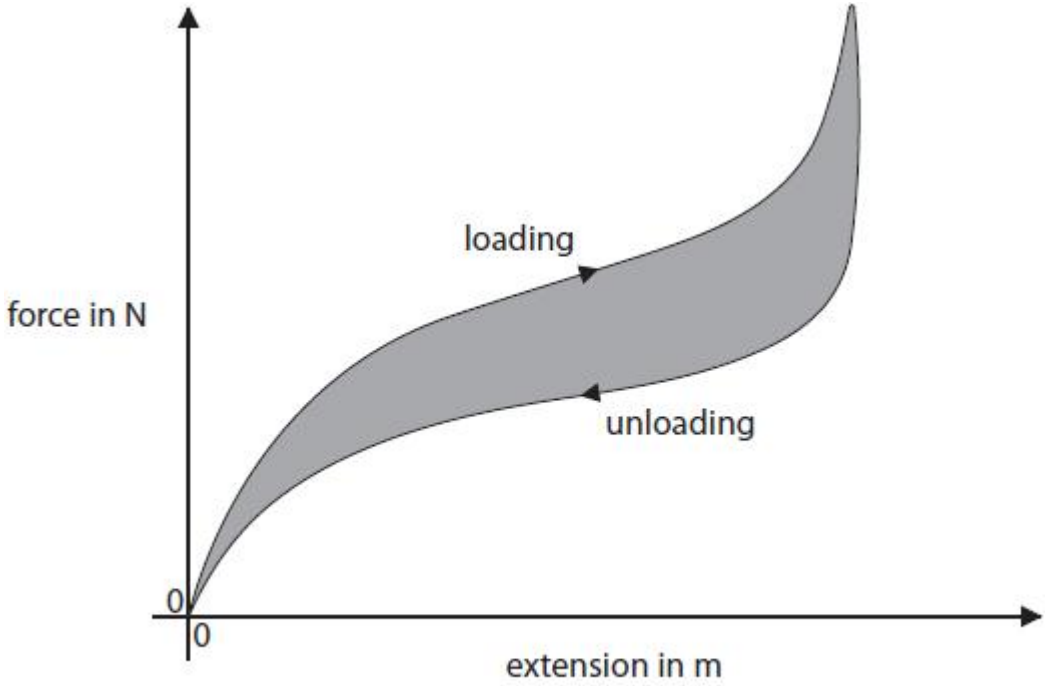

Figure 16

Explain how the shape of this graph shows that the distortion of the piece of rubber being stretched is different from the distortion of a spring being stretched.

(2)

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(Total for question = 4 marks)

Q7.

This question is about pressure.

Figure 7 shows the atmospheric pressure at different heights above the Earth's surface.

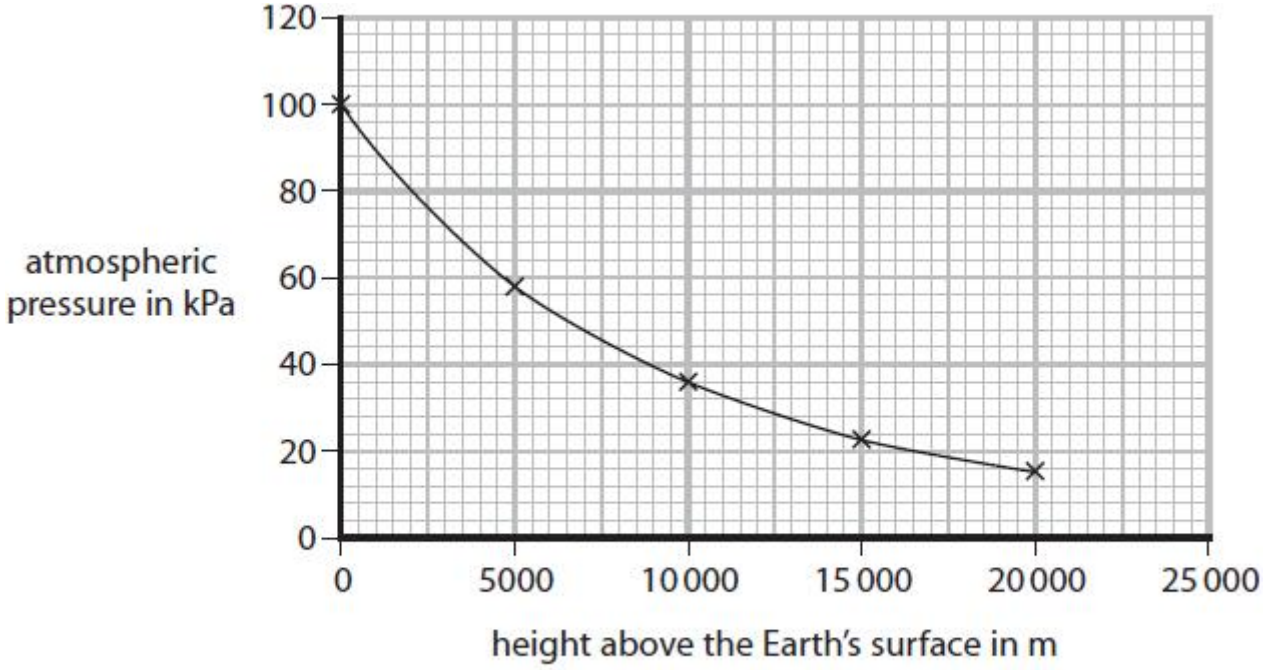

Figure 7

(i) Describe how the atmospheric pressure changes with height above the Earth's surface.
Use data from Figure 7 to support your answer.

(3)

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(ii) Suggest **one** reason why the atmospheric pressure changes with height above the Earth's surface.

(1)

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(Total for question = 4 marks)

Q8.

A student carries out an investigation to show how pressure varies with depth in water.

A pressure sensor is attached to a rule.

The rule and pressure sensor are lowered into the water in a tank, as shown in Figure 8.

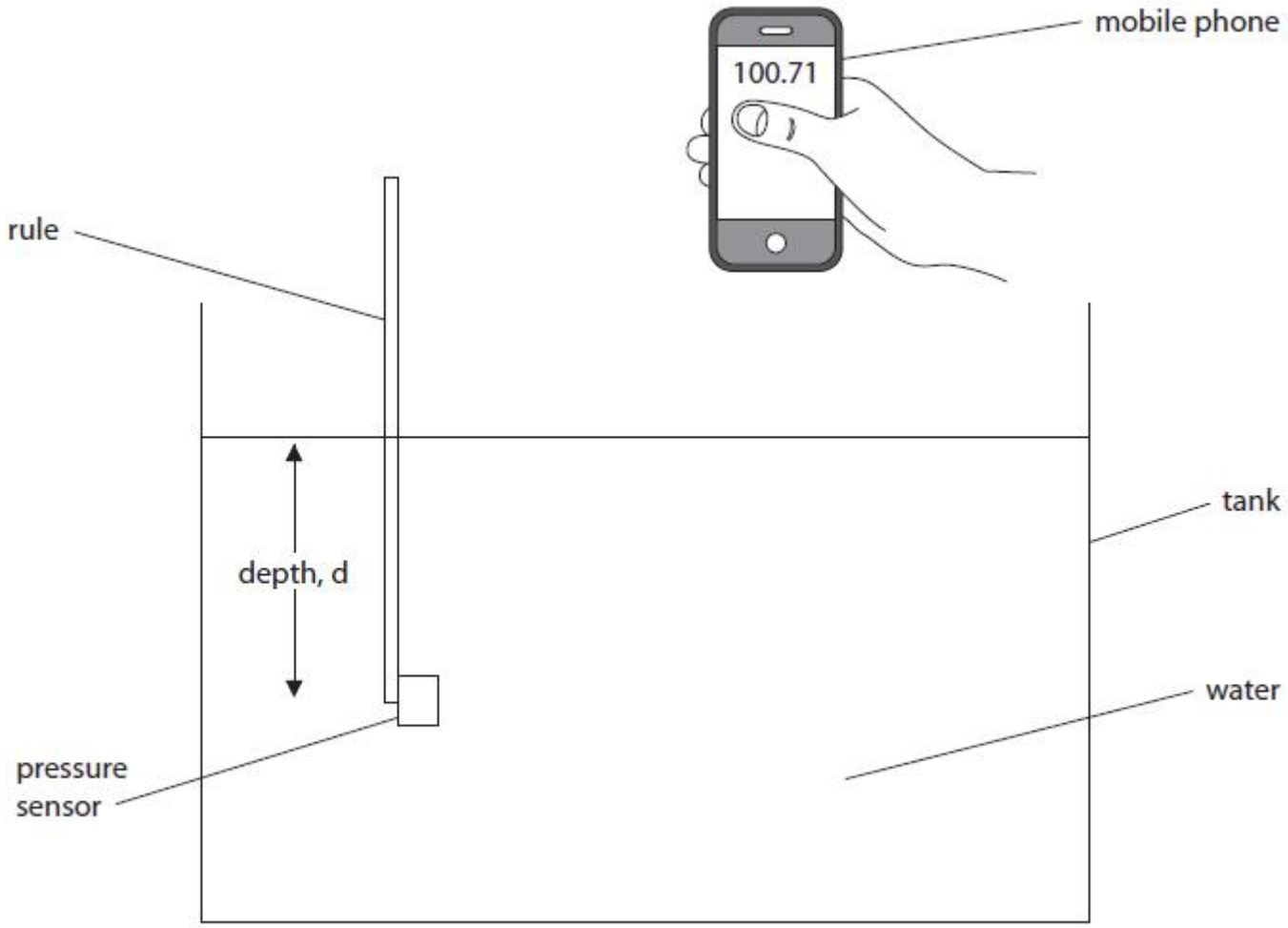

Figure 8

The depth of the pressure sensor below the surface of the water is read from the scale on the rule.

The pressure is displayed on the student's mobile phone which receives a signal from the pressure sensor.

Figure 9 gives some of the readings.

depth in m	pressure in kPa
0.050	99.15
0.100	99.70
0.150	100.15
0.200	100.70
0.250	101.15
0.300	101.70

Figure 9

Figure 10 shows a graph with some of the results plotted, but two of the points are missing.

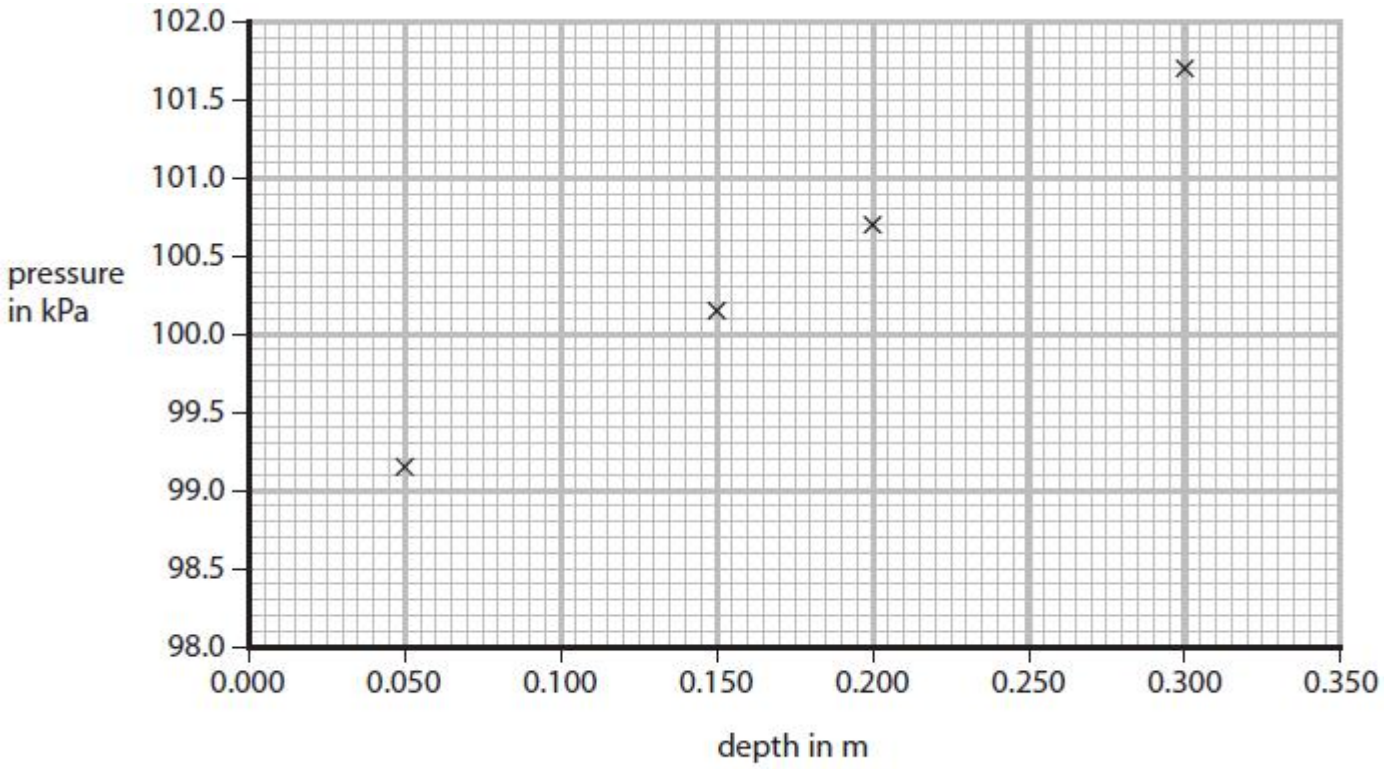

Figure 10

The student repeats the investigation using seawater and draws a graph of the results.

The seawater is more dense than the water used previously.

Compare the graph for seawater with the graph in Figure 10.

(2)

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(Total for question = 2 marks)

Q9.

Figure 17 shows information about the pressures in the ocean and in the atmosphere of a distant planet.

Graph A shows the variation of pressure as the depth in the ocean increases.

Graph B shows the variation of pressure as the height in the atmosphere increases.

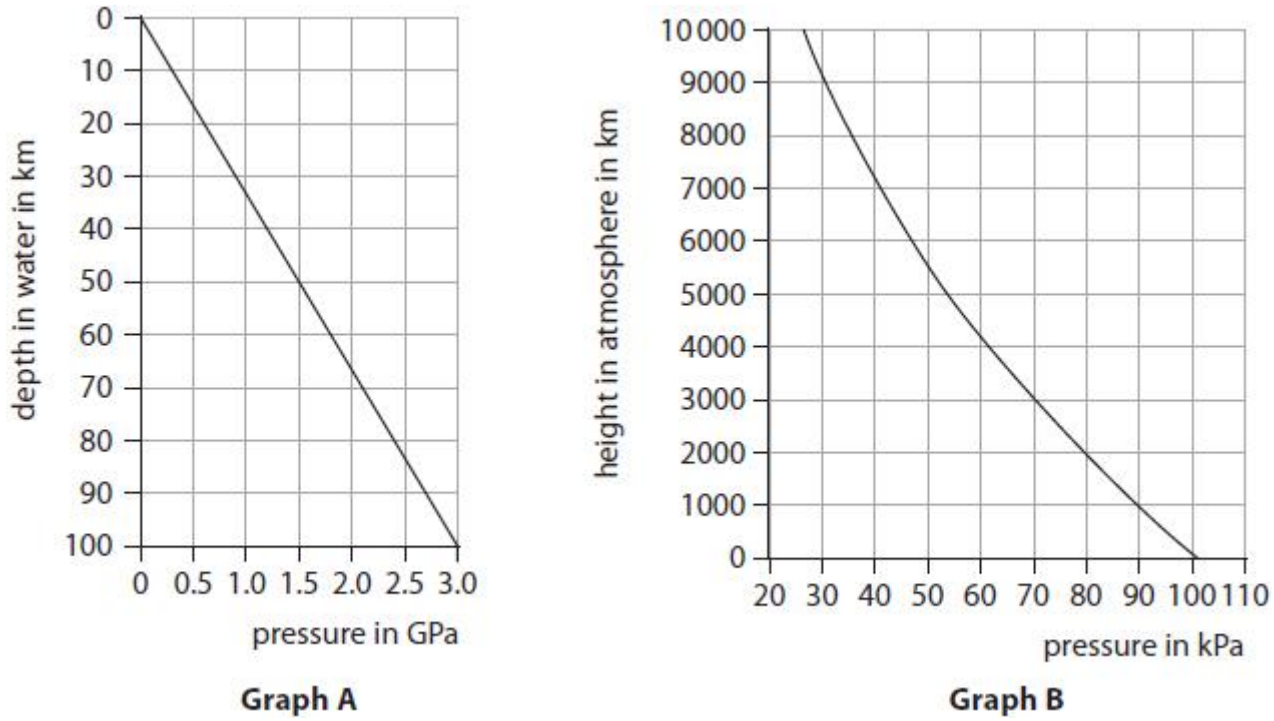

Figure 17

(i) Use information from Graph A to obtain a value for the density of the ocean water.
Use an equation selected from the list of equations at the end of this paper.

(4)

density = ... kg/m^3

* (ii) Explain the similarities and differences in the pressure variations shown in graphs A and B.
Your answer should refer to both the particle model (kinetic theory) and to density.

(6)

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(Total for question = 10 marks)

Q10.

Figure 11 shows how atmospheric pressure changes with height above sea level.

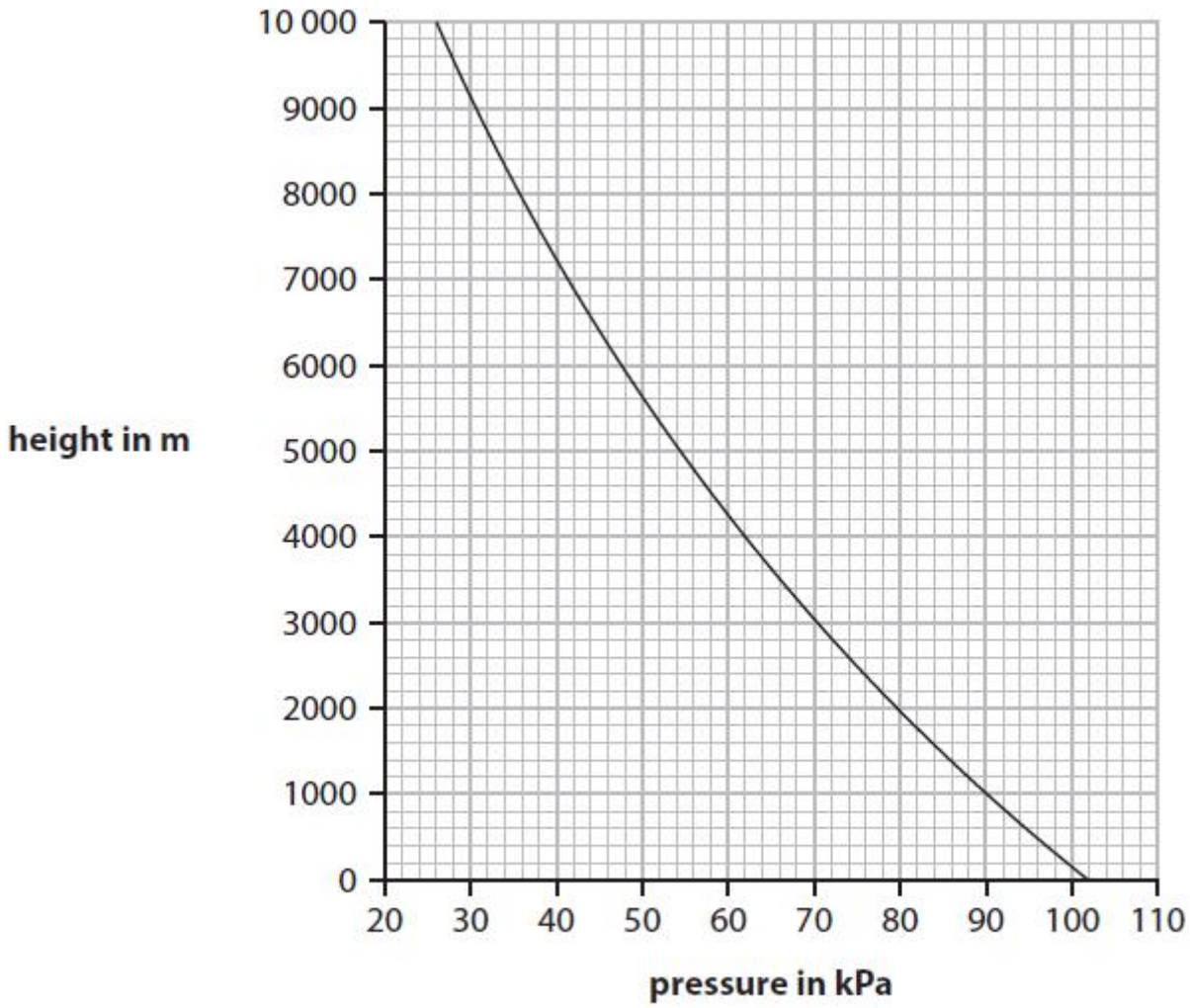

Figure 11

(i) Using the graph, describe how atmospheric pressure changes with height above sea level.

(2)

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(ii) The top of Mount Everest is 8850 m above sea level.

Using the graph, estimate the atmospheric pressure at the top of Mount Everest.

(1)

pressure = ... kPa

(iii) On a different day, the pressure at sea level is 104 kPa and the pressure at a height of 2500 m is 74 kPa.

Calculate the percentage change in pressure from sea level to the height of 2500 m.

(2)

percentage change = ... %

(Total for question = 5 marks)

Q11.

Figure 20 shows a metal block being weighed by a newton meter (force meter).

In Figure 20a the block is hanging in a liquid.
In Figure 20b the **same** block is hanging in air.

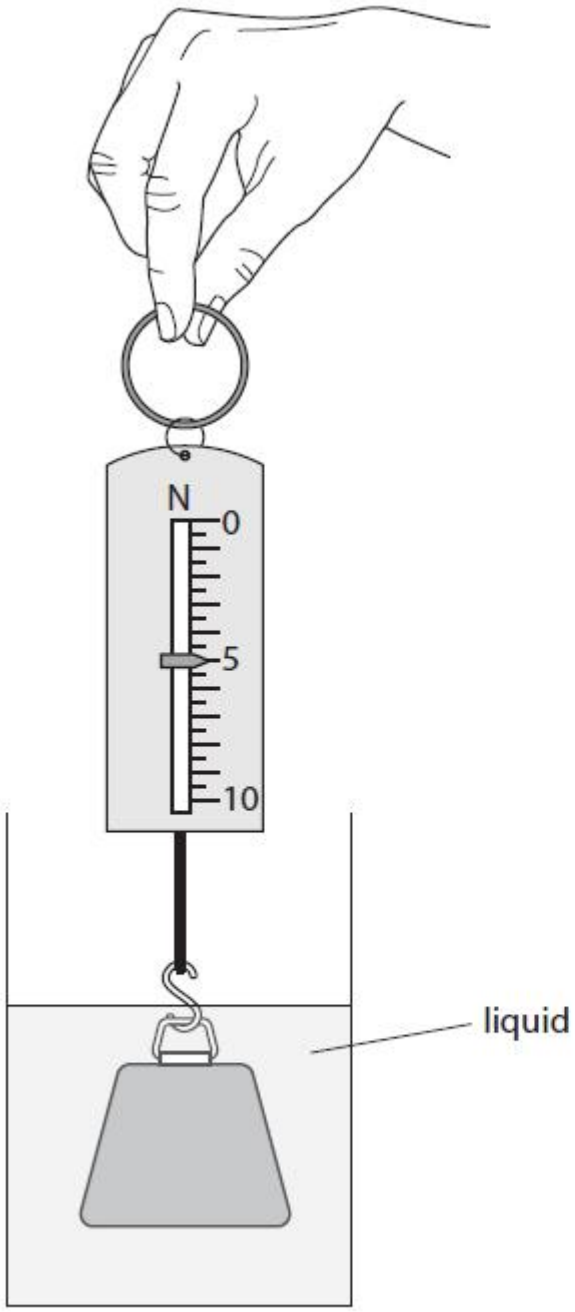

Figure 20a

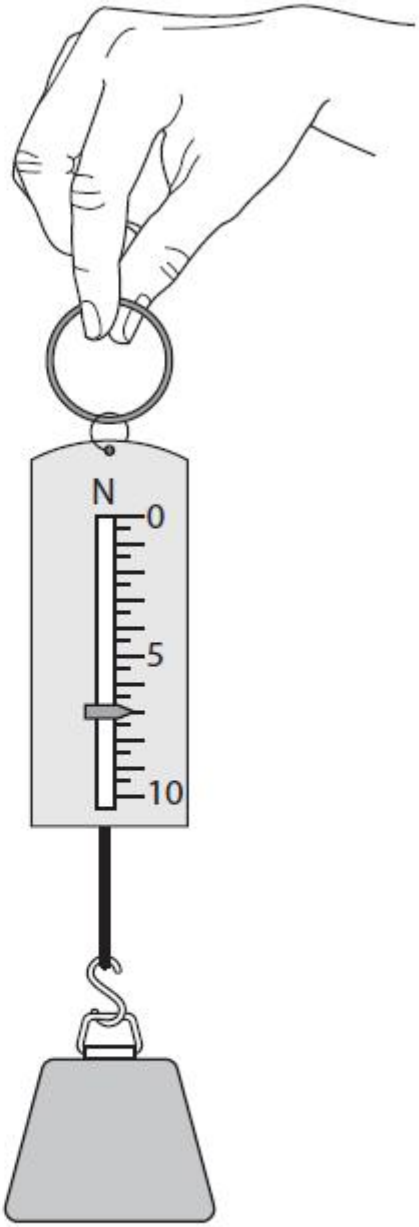

Figure 20b

(i) Explain why the reading on the newton meter is different in Figures 20a and 20b.

(2)

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(ii) The density of the liquid is 1.4×10^3 kg/m^3.

Calculate the density of the block.
Gravitational field strength, *g*, = 10 N/kg

(4)

density = ... kg/m^3

(Total for question = 6 marks)

Mark Scheme

Q1.

Question number	Indicative content	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative (example) content below is not prescriptive and candidates are not required to include the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. AO2 **Pressure** • difference in pressure between top and bottom of boat • top pressure is atmospheric • pressure on bottom of boat atmospheric plus that due to depth of water. **Unloaded boat** • density of boat less than density of water • floating objects are partially immersed • floating objects displace fluid / water • upthrust is due to the difference in pressure • upthrust is equal to the weight of the boat • upthrust is equal to the weight of fluid / water displaced **Boat with load** • the weight/density of the boat increases because of the load added • more upthrust is needed to balance the extra weight of the boat • more water has to be displaced to provide the upthrust • when the boat floats lower in the water it displaces more water • the weight of water displaced is the upthrust and is equal to the weight of the boat	**(6)**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. Lines of reasoning are unsupported or unclear. (AO2)
Level 2	3–4	The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. Lines of reasoning mostly supported through the application of relevant evidence. (AO2)
Level 3	5–6	The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. Lines of reasoning are supported by sustained application of relevant evidence. (AO2)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Elements of physics present i.e. isolated knowledge of principles **two** unconnected statements from any section	Possible candidate responses pressure difference upthrust water displaced displacement floating
Level 2	3–4	Additional guidance Some knowledge of principles with a logical connection made in one section and statement from one other section	Possible candidate responses upthrust and weight are balanced /upthrust is equal to the weight of the boat when load added upthrust increases difference in pressure between the top and bottom of the boat
Level 3	5–6	Additional guidance Detailed knowledge of principles with logical connections made in two sections.	Possible candidate responses upthrust is equal to the weight of water displaced. when load is added, weight increases more water is displaced

Q2.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 (3 marks)** • upthrust is the force on the submarine in the water (submerged) in a fluid • upthrust on the submarine and its weight act in opposite directions • upthrust is equal to the weight of water displaced by the submarine • the difference in pressures on the upper and lower surfaces of the submarine causes the upthrust **AO2 (3 marks)** • the volume of the submarine is fixed so the upthrust on the submarine is constant • increasing/decreasing volume of water in tanks increases/ decreases weight of submarine but does not affect upthrust • if weight increases to become greater than upthrust there is a resultant downward force on the submarine so the submarine sinks • if weight decreases to become less than upthrust there is a resultant upward force on the submarine so the submarine rises	**(6)**

Level	Mark	Descriptor
	0	No awardable content.
Level 1	1–2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3–4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5–6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Q3.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification via a judgement (1 mark) to reach a conclusion via justification/reasoning (2 marks): • pressure in A is the highest and pressure in C is the lowest (pressure in B is between them) (1) • pressure depends on depth of liquid (so) can compare A and C because same liquid (hence) pressure in A is twice that of C (1) • pressure depends on density of liquid (so) can compare A and B since same depth hence pressure in A greater than pressure in B (1)	allow a mathematical approach, i.e. calculating all three pressures from the relevant data	**(3)**

Q4.

Question number	Answer	Additional guidance	Mark
		ignore gravity accept reverse arguments	(2) AO3.2
	an explanation linking: density decreases as height increases (1) with (because) particles are further apart (higher up) (1)	 density decreases as you go higher accept fewer particles per unit volume accept particles more spaced out	

Q5.

Question number	Answer	Additional guidance	Mark
	an explanation linking the area (of contact between person and bed) is smaller when standing up (1) same weight (over smaller area) so the pressure is greater when standing up (1)	accept reverse arguments weight is more concentrated / not distributed /not spread across bed (when standing up) uses p = F/A argument (as a consequence of the smaller area, pressure is bigger)	(2) AO2

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	A description including any one from the following (1) **measure** a length or a specific distance related to the rubber or weights on a hanger OR with a **named device** (e.g. metre rule / stick / ruler / measuring tape) OR note position of a fixed point on rubber / weight carrier **AND** extension calculated / measured as the change in or difference between two positions or lengths or extensions (1)	evidence may be taken from additions to the diagram ignore vague statements such as see how it much it extends	(2)

Question Number	Answer	Additional guidance	Mark
(ii)	An explanation linking graph of rubber band is non-linear / curved / not directly proportional (1) graph for unloading does not go through same points as loading (1)	 (graph for) spring would be straight (graph for) spring would only have one line / go through the same points ignore reference to returning to original shape / length	(2)

Q7.

Question number	Answer	Additional guidance	Mark
(i)	increase in height results in decrease in pressure (1)	pressure decreases with height accept inversely proportional in this context accept negative correlation	(3) AO3
	non-linear relationship (1)	double the height does not result in half the pressure pressure not does change evenly description of graph e.g. curved / not straight	
	use of numerical data (1) at least two different pressure and height values from the graph	calculation of change in pressure e.g. 5000m to 10000 m pressure went down by 22	

Question number	Answer	Additional guidance	Mark
(ii)	any **one** from	accept oxygen / atmosphere for air	**(1) AO1**
	air becomes less dense (1)	air gets thinner / (air) particles further apart / fewer particles / less particles	
	smaller weight (of air) above (1)	less air above / smaller height of air above	
	lower temperature (1)	ignore change in value of g with height	

Q8.

Question number	Answer	Additional guidance	Mark
	any two from	**credit mark points seen on graph**	**(2) AO3.2**
	pressure(s) would be greater (values) (1)		
	steeper gradient of graph (1)	bigger gradient / steeper line (of best fit)	
	both straight lines (1)	both linear	
	intercept (on pressure axis) the same (1)	pressure at surface is the same	

Q9.

Question number	Answer	Additional guidance	Mark
(i)	rearrange $p = \rho g h$ to give $\rho = p / (g h)$ (1) substitution using any point from graph (1) e.g. depth = 50km and pressure = 1.5 GPa $\rho = p / (g h)$ $= 1.5 \times 10^9 / (10 \times 50 \times 10^3)$ Evaluation (2) $= 3000$ (kg/m^3)	rearrangement and substitution in any order allow any combination from the graph and ignore 'pot' error here 'pot' error scores 2 marks maximum	(4)

Question number	Indicative content	Mark
(ii)	Answers will be credited according to the candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all of the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO2 (3 marks) A03 (3 marks)** **A03 Interpretation and evaluation from the graph** Similarities: • both show increasing pressure with depth • both show a range of pressures over kilometre depths / heights Differences: • ocean water shows a linear relationship (straight line) but atmosphere gives a non-linear (curved) relationship • density of ocean water not changing with depth but density of atmosphere changes as you go higher • The pressures in the ocean recorded are much bigger (GPA compared with kPa) The depth of the ocean shown is up to 100km whereas the height of the atmosphere involved is only up to 10km **A02 Link between graph shapes and underlying physics** Similarities: • pressure is due to (increasing) weight of fluid (liquid / gas) above • more molecules above Differences: • atmosphere becomes thinner the higher you go molecules in the ocean stay (on average) the same distance apart but in the atmosphere they get further apart (on average) as you go higher up	(6)

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1–2	• Interpretation and evaluation of the information attempted but will be limited with a focus on mainly just one variable. Demonstrates limited synthesis of understanding. (AO3) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3–4	• Interpretation and evaluation of the information on both variables, synthesising mostly relevant understanding. (AO3) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5–6	• Interpretation and evaluation of the information, demonstrating throughout the skills of synthesising relevant understanding. (AO3) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Q10.

Question number	Answer	Additional guidance	Mark
(i)	a description including pressure increases as height decreases (1) OR as height increases pressure decreases (1)	negative correlation	(2) AO3.2

	non-linear (1)	implication of non-linear e.g. curved OR not proportional OR gradient increases as height increases OR gradient decreases as pressure increases	

Question number	Answer	Additional guidance	Mark
(ii)	accept any answer from 30 to 34 (kPa) (1)		(1) AO3.2

Question number	Answer	Additional guidance	Mark
(iii)	substitute into % calculation (1) $\frac{74 - 104}{104}$ (x 100) evaluation (1) (-) 29 (%)	 $\frac{104 - 74}{104}$ (x 100) any number rounding to (-)29(%) e.g. (-)28.8(%) award full marks for the correct answer without working award 1 mark for (-)0.29 OR (-)0.288 award 1 mark for (-)40(.54) (%) or (-)41 (%) award 1 mark for 71 (%)	(2) AO2.1

Q11.

Question number	Answer	Additional guidance	Mark
(i)	an explanation linking (apparent) loss in weight of block in liquid(1) (equal to) weight of liquid displaced (1)	 upthrust /buoyant force on block liquid more dense than air	(2) AO1

Question number	Answer	Additional guidance	Mark
(ii)	calculation of weight lost (1) 2 (N) attempt to determine volume of block (1) (V=) 1.43(x 10^{-4}) substitution to determine density of block (1) $(\rho =) \frac{7 (x 10^{-1})}{1.43 (x10^{-4})}$ evaluation (1) (ρ =) 4900 (kg/m^3)	accept (7 – 5) accept 0.2 (kg) accept $(V =) \frac{2 (x 10^{-1})}{1.4 (x 10^{3})}$ or $(V =) \frac{5 (x10^{-1})}{1.4 (x10^{3})}$ or 3.6 (x10^{-4}) (weight of 5 N used) accept $\frac{7 (x10^{-1})}{3.6 (x 10^{-4})}$ accept values that round to 4900 e.g. 4895 4900 to any other power of ten scores 3 marks (incorrect unit conversion) 1960 scores 3 marks 1960 to any other power of ten scores 2 marks (used wrong weight of liquid) 100 000 scores 3 marks 100 000 to any other power of ten scores 2 marks	**(4)** **AO2**

		(wrong rearrangement) 250 000 scores 2 marks 250 000 to any other power of ten scores 1 mark (used wrong weight of liquid and wrong rearrangement) award full marks for the correct answer without working	